



WASHINGTON STATE
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Injury risk factors and prevention strategies for Ironman triathlons: The athletes' perspectives

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Objectives

- Triathlon injury background
- TriSafe Research Initiative
- Qualitative research methodology
- Results
- Discussion
- Future Directions



Background

- Annually: 300k+ competitive triathletes in US; 200k Ironman competitors globally
- Recreational triathletes train ~8–16 h per week; elite triathletes train ~16–20 h per week

	Swim (m)	Cycle (km)	Run (km)	Time to completion
Sprint	750	20	5	~1–2 hours
Olympic	1,500	40	10	~2.5–3.5 hours
Half-Ironman	~1,800	90	21.1	~3.5–7 hours
Full Ironman	~3,860	~180.2	42.2	~8.5–17 hours
Ultra-triathlon	38,600	~1,800	422	Multiple days

Injury Rates: Triathlon vs. Team Sports

- ~37% to 91% of triathletes report sustaining an injury in a year (average 2.9 injuries per athlete)

Athlete Population	Overall	Training	Competition	Reference
Triathlon	--	0.7–5.4	9–20	Feletti et al. (2022)
*Football	4.6–9.9	1.3–4.7	13.9–91.8	Zech et al. (2021)
*Volleyball	2.4–3.8	--	4.4–12.2	Zech et al. (2021)
Rugby	17.7–45.0	--	16.9–55.6	Zech et al. (2021)
Handball	--	2.1–4.1	13.0–113.0	Zech et al. (2021)

*Higher IR for female athletes

Triathlon-related Deaths

- While rare, the mortality rate during triathlon competitions is 1.5 per 1000 participants (twice that of marathons)
- Triathlon-related death primarily occurs during the swim (drowning, cardiac arrest) or during cycling (blunt force trauma via collision)
- Reviews of these fatalities found no clear links to race length, type of swim venue, swim start method, triathlete experience level, swimming ability, medical exam, or underlying medical conditions
 - Available autopsy reports did find a portion of cardiac abnormalities (44%) in the deceased

The Gaps

There is a lack of triathlon-specific evidence-based guidance regarding injury risk & injury prevention

- Triathlon has received far less attention from researchers, compared to swimming, cycling, or running
- Few studies have investigated triathlon-specific injury risk factors
 - Focus has been on injury surveillance, etiology, and mechanism of injury
 - Injury data is often compiled from different race lengths
- The underlying risk factors elevating the risk of serious injury and death during triathlon events remain unclear

TriSafe Research Initiative

TriSafe Purpose: To investigate active triathletes' perspectives on triathlon-specific injury risk factors and injury prevention strategies to be used as targets for future injury prevention efforts

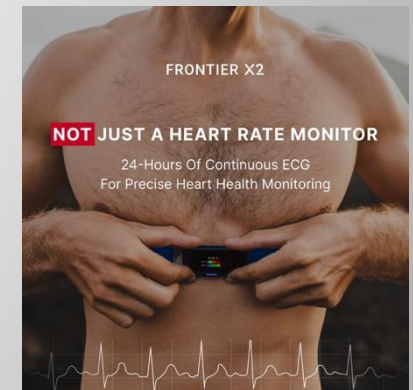
- **Triathlon Risk Surveillance and Assessment of Fatalities and Environments (TriSafe)**
 - Data collection period: 7 days pre-race, during race, 7 days post-race
 - ✓ Biometric data: Oura Ring (sleep), Polar watch (activity, HR, HRV), Fourth Frontier chest strap (HR, HRV)
 - ✓ Surveys: daily, weekly, pre-post race
 - ✓ **Optional focus groups and interviews post race (our focus today)**



Polar heart rate watch



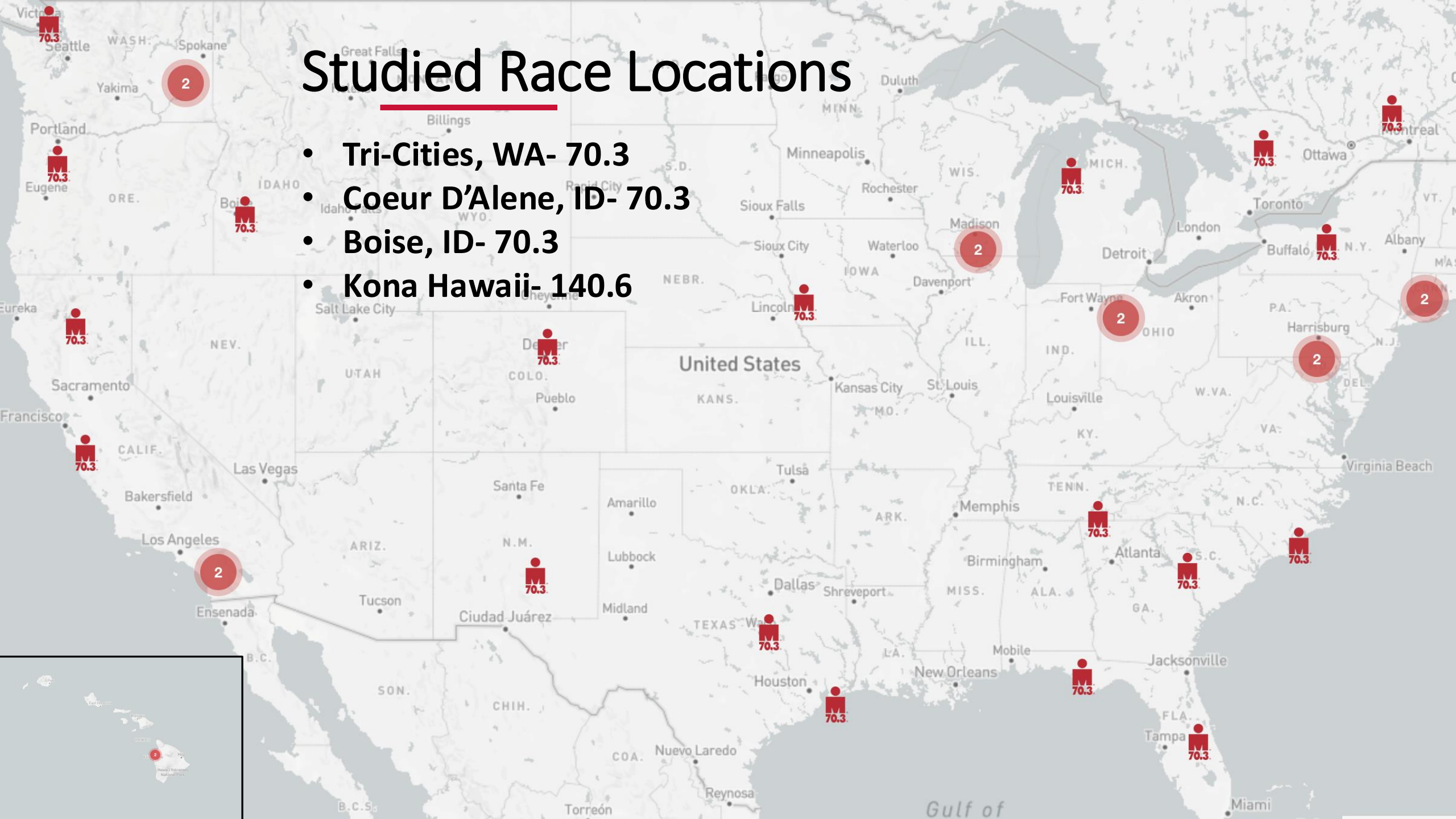
Oura Ring



Fourth Frontier

Studied Race Locations

- Tri-Cities, WA- 70.3
- Coeur D'Alene, ID- 70.3
- Boise, ID- 70.3
- Kona Hawaii- 140.6



Qualitative Methods



Qualitative Research Methodology

Like quantitative research, qualitative research is an umbrella term with different approaches

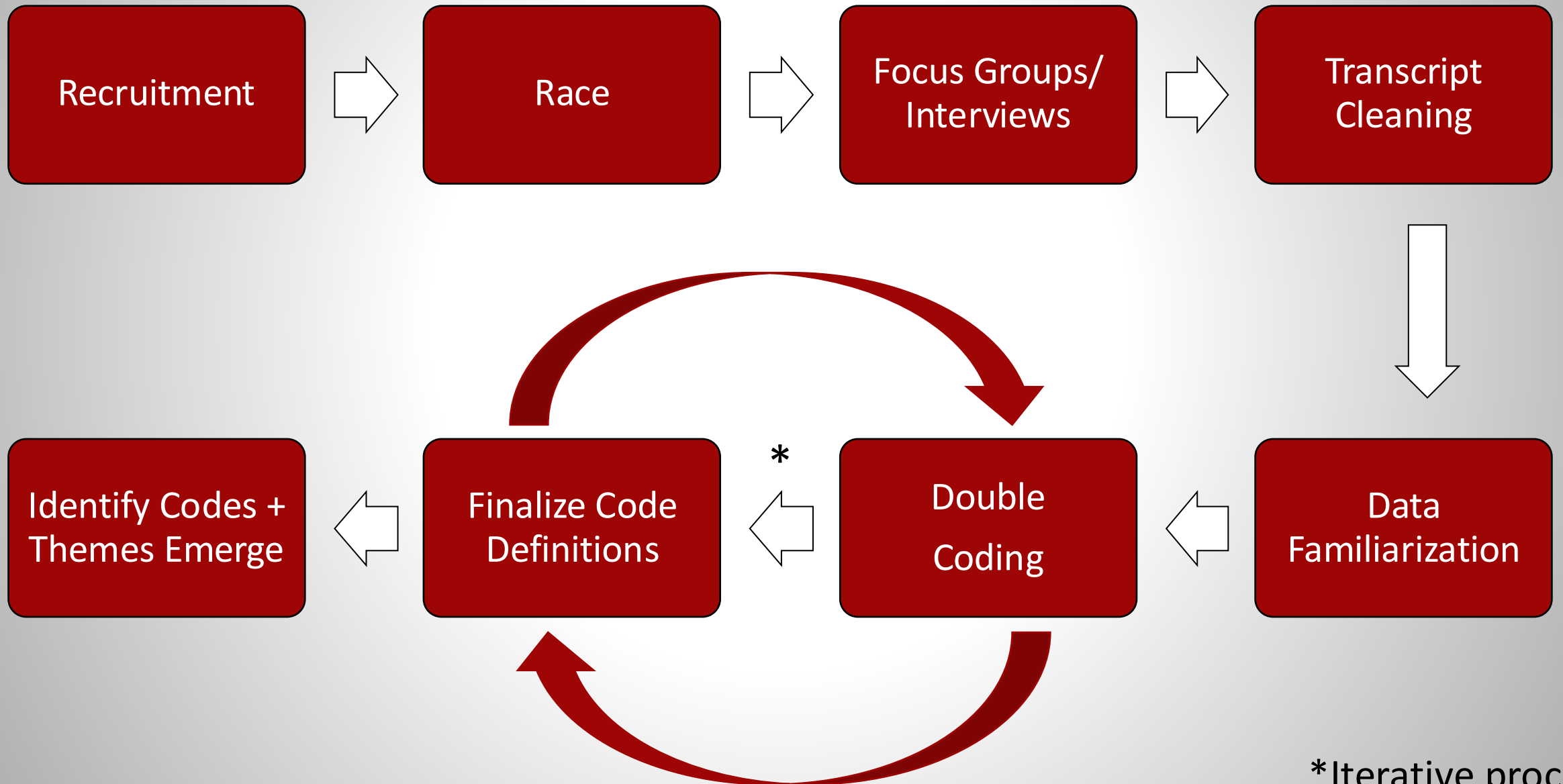
Qualitative studies focus on:

- **Ensuring the voices of participants are heard** in the studies
- Allowing **codes and themes to emerge from the data**
- Including the **context** of “real life” behavior
- Understanding the “**why**” and “**how**”
- **Offering deeper insights** into behaviors
 - Specific to triathlons, the insights also include race challenges and safety impacts
- **Methods for our study:**
 - ✓ Data was collected through **focus groups** and **interviews**
 - ✓ For our analysis we used combined **latent content analysis with summative analysis**



Focus Group and Interview Questions

1. **What do you think are the most important factors to consider when assessing (physical) risk before, during, and after the Ironman?**
2. How do you mentally prepare for the race? What is the hardest part mentally about the Ironman?
3. How was device function? How easy were they to use?
4. How was study communication?
5. **Did you have any concerns about health and injury during the race? If so, what did you do (e.g., no change in effort, backed off, got assistance, etc.)**
6. What else should we know about to inform future studies?



*Iterative process

Results



Demographics & Major Themes

	Males (n=14)	Females (n=8)	All (n=22)
Age (y)	49 (23–69)	48 (23–64)	49 (23–69)
Height (m)	1.79 (1.68–1.88)	1.73 (1.60–1.88)	1.76 (1.60–1.88)
Weight (kg)	79.61 (65.91–102.27)	61.17 (52.27–84.09)	74.36 (52.27–102.27)
Triathlons Completed	9 (1–22)	13 (1–80)	10 (1–80)

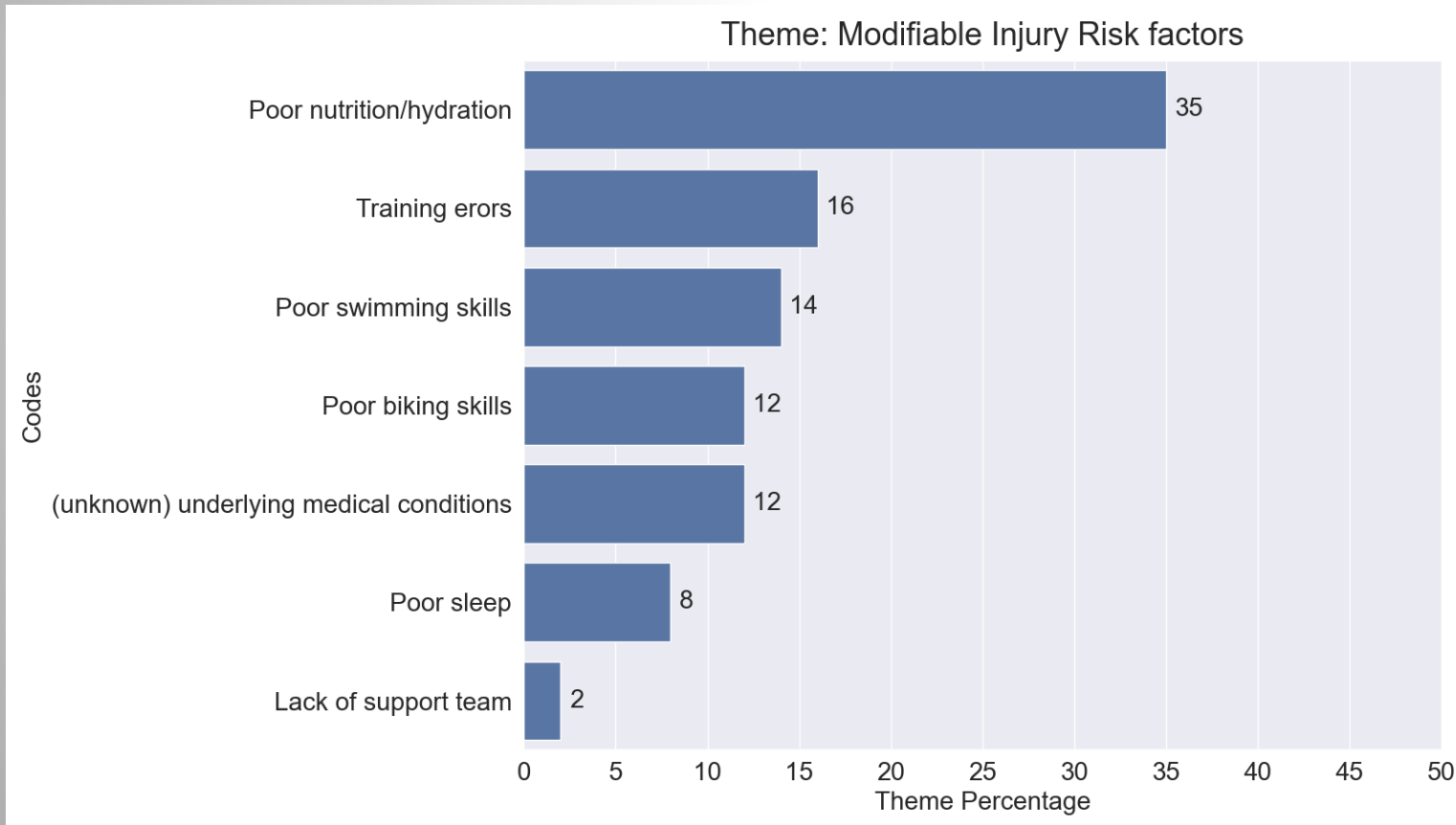
Injury Risk Factor Themes:

- Non-modifiable Risk Factors
- Modifiable Risk Factors
- Acute Risk Factors
- Warning Signs (“Red Flags”)

Injury Prevention Strategies Themes:

- Pre-race Strategies
- During Race Strategies
- Post-race Strategies

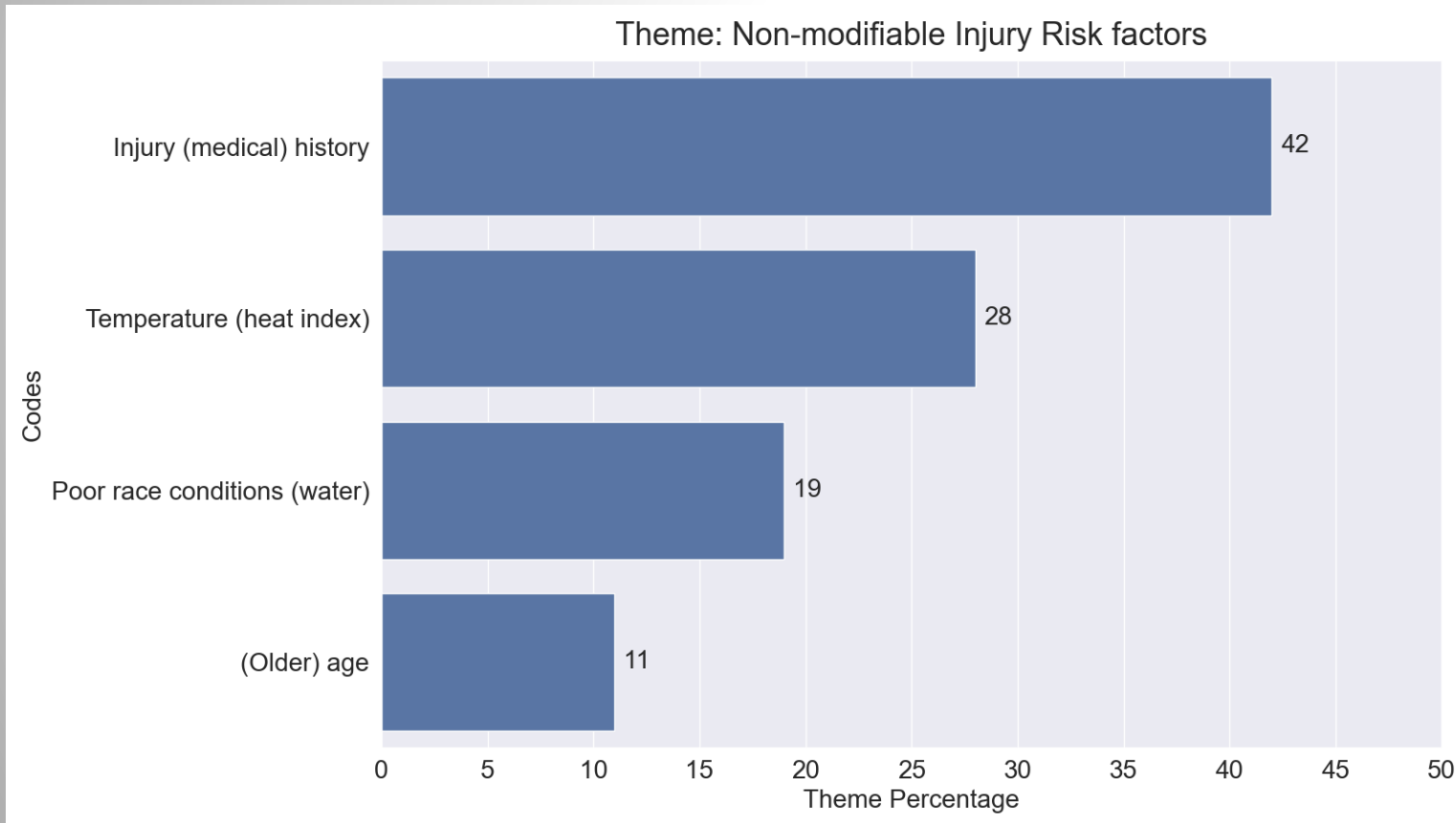
Modifiable Risk Factors



“The nutrition; just had to stick to the plan to avoid any issues from that. And I'm really conscious of like safety on the bike.”

“If you aren't a good bike rider that can handle the bike in a crowd. Then you should probably look for a different race.”

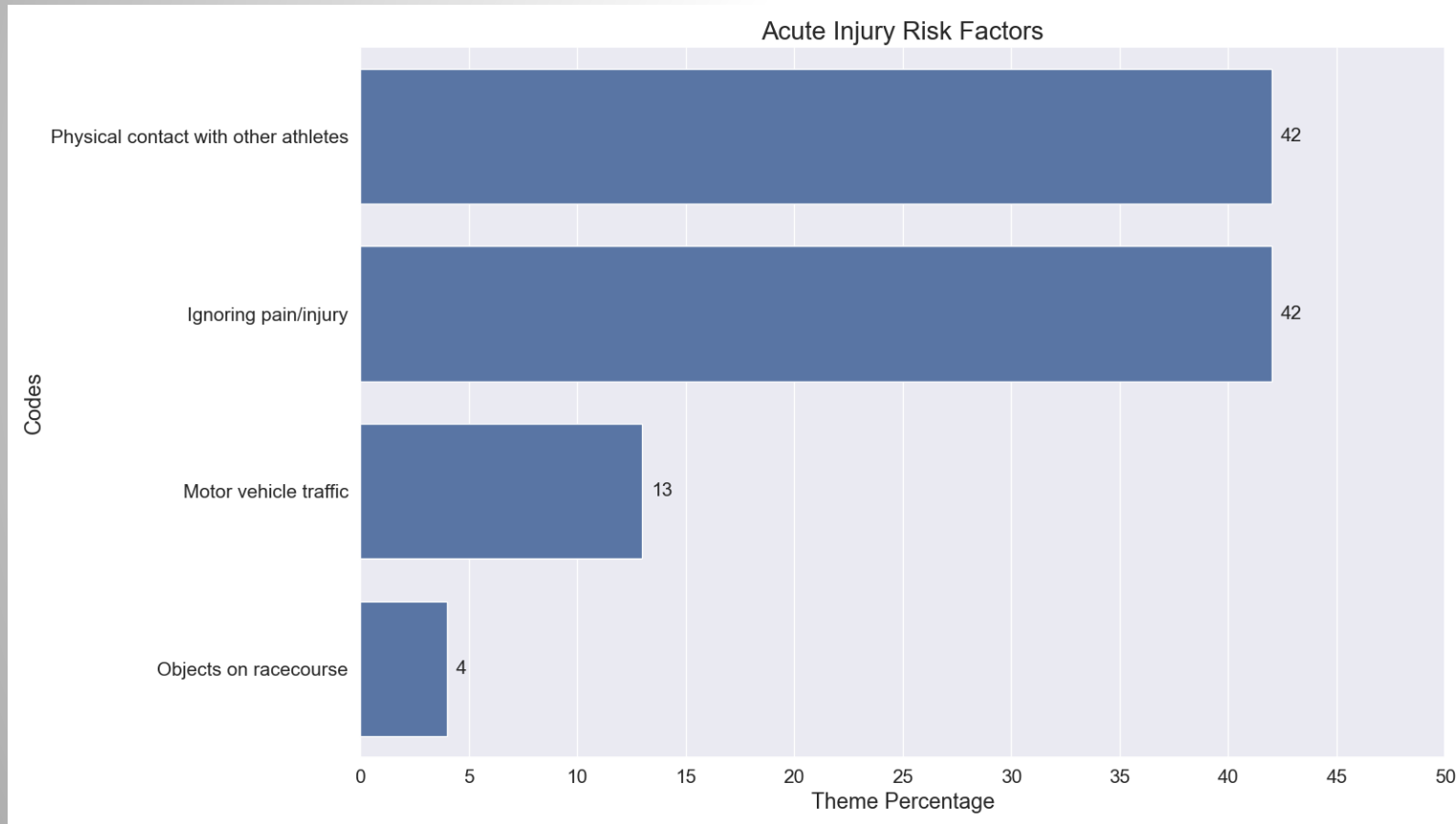
Non-modifiable Risk Factors



“I’d say pre-existing conditions, or is there something that might predispose me to having a problem...”

“As I’ve aged up, I’ve become a real ‘heat weenie’.”

Acute Risk Factors

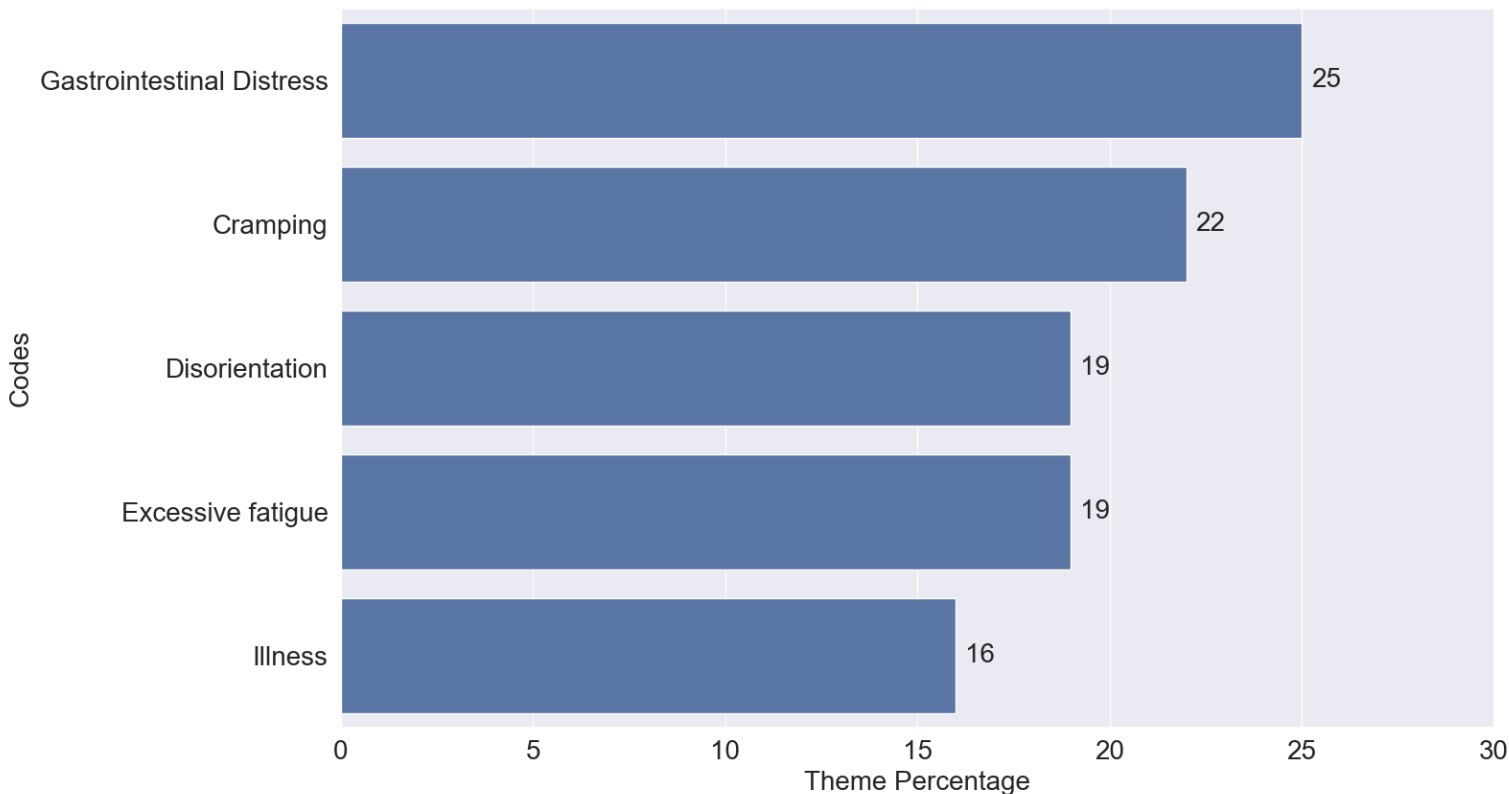


“Those more like overuse injuries, I think, can be really common from the run, where somebody has like a nagging pain for 4 hours, and then all of a sudden, it's like a 6-month long recovery process.”

“So, you know you're swimming, and somebody kicks you in the head, or you know things like that you cannot control.”

Warning Signs (“Red Flags”)

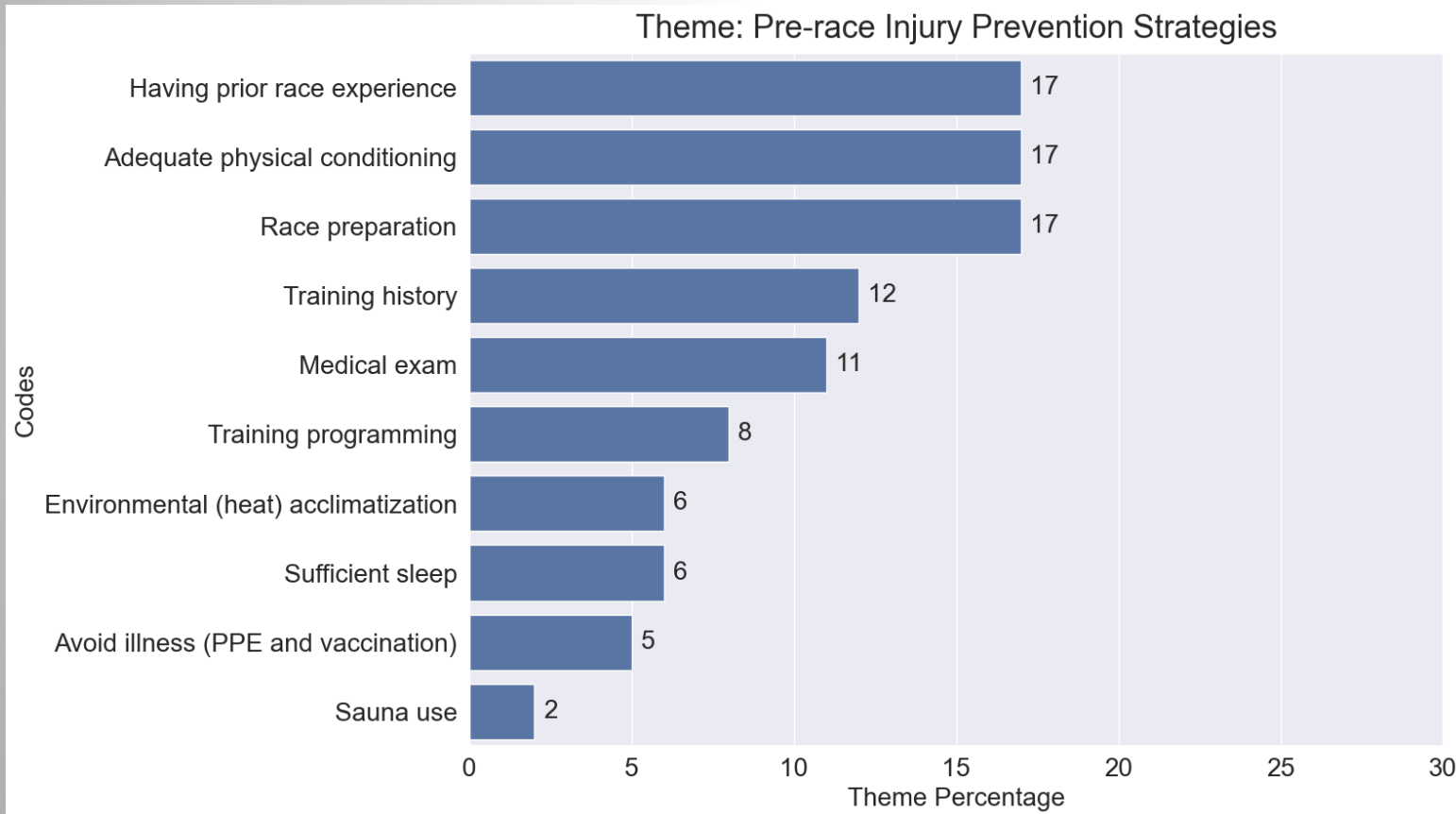
Theme: Injury Warning Signs (Red Flags)



“If you’re feeling an extreme amount of fatigue and maybe like trying to use your gels or like your nutrition and that’s not helping at all. I think that would be kind of a red flag.”

“If it's hot and humid...my GI [gastrointestinal] system just like locks up.”

Pre-race Injury Prevention Strategies

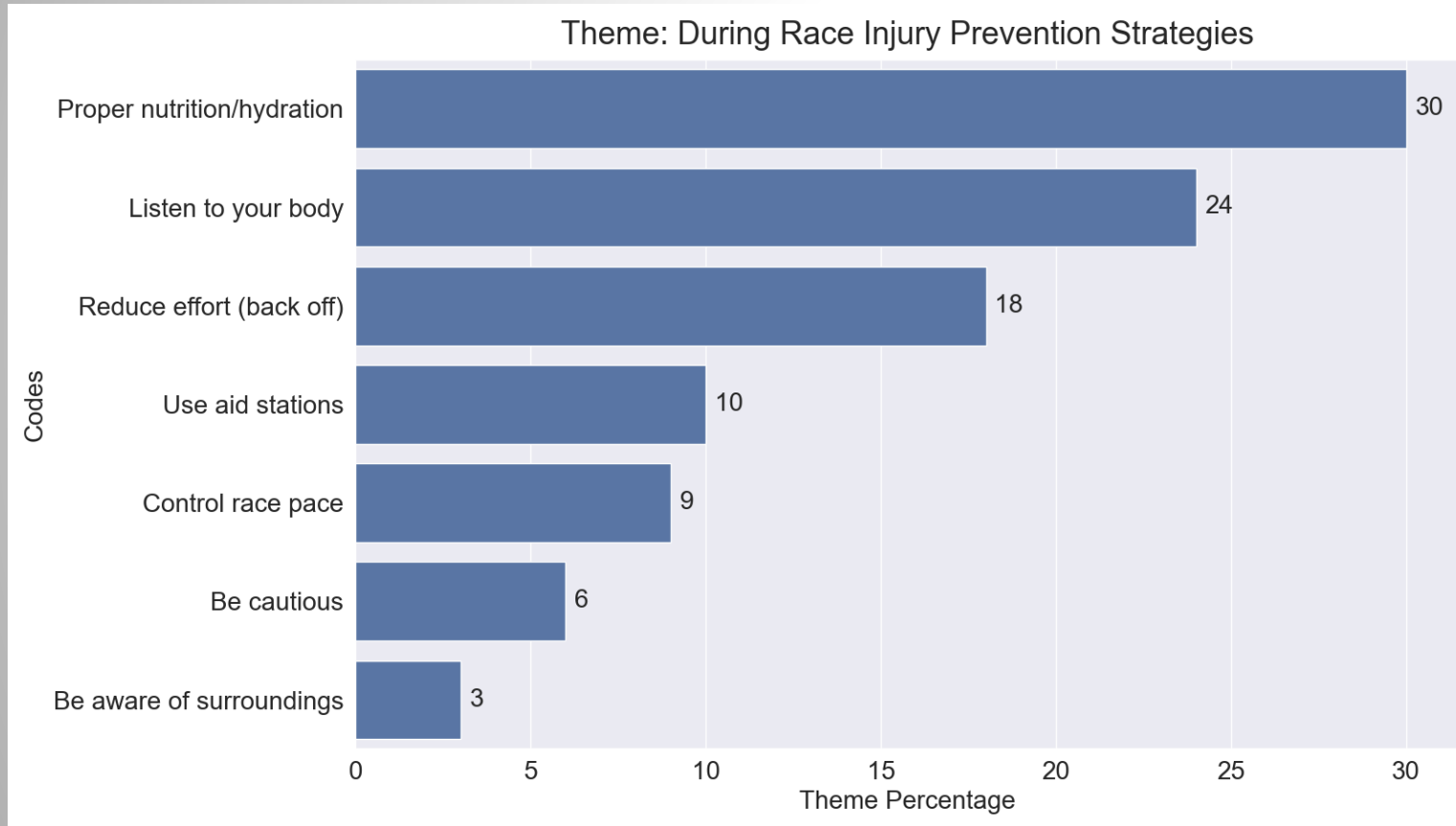


“To know that you can compete in these activities for the three consecutive sports and that you have trained adequately for them.”

“If you have experience doing this kind of thing, it’s helpful to know that training load you’re working up to.”

“Two to three nights before, it’s about sleep and nutrition.”

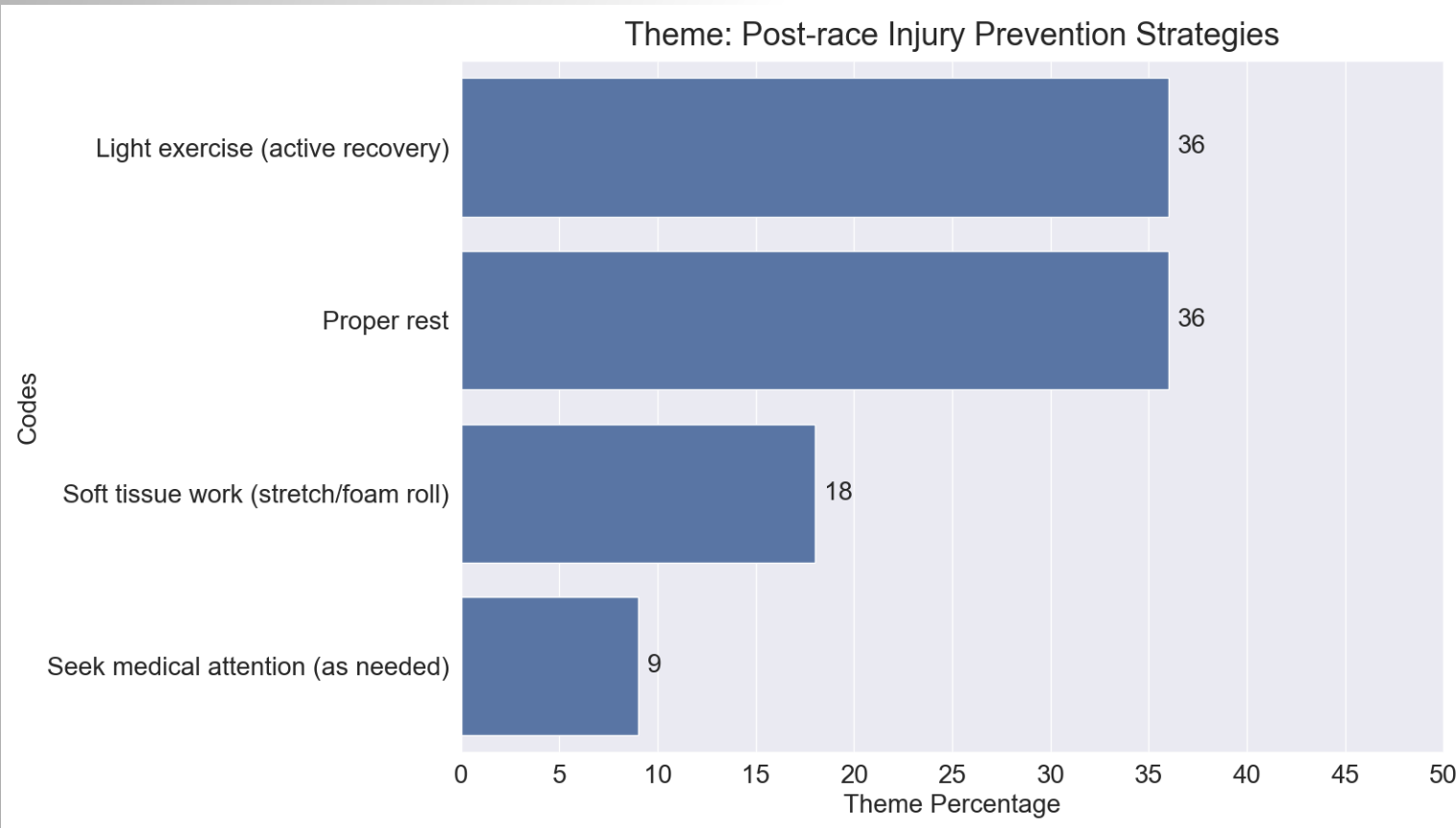
During Race Injury Prevention Strategies



“Make sure your hydration, your nutrition plan is right and you’re keeping a straight head.”

“If I started feeling like a tweak or a cramp, or like something where I was like oh, that's like hurting more than like a fatigue baseline just like, take backing it off like just a tiny bit.”

Post-race Injury Prevention Strategies



“Try and stretch out whenever I get home and going to foam roll, but I’m keeping it very low key.”

“Making sure you’re getting some rest and not getting back into your training plan again too quickly to make sure you’re taking care of all the aches and pains.”

Discussion



Translating Research into Injury Prevention Practice*

TRIPP Stages:

1. Injury surveillance
- 2. Injury etiology and mechanism of injury**
- 3. Identifying and developing preventive measures**
4. “Ideal conditions” / scientific evaluation
- 5. Intervention context to inform implementation strategies**
6. (Re)evaluate effectiveness of preventive measures in implementation context

Discussion: Key Takeaways

Modifiable Risk Factors Theme:

- **Swimming and cycling proficiency**—along with **adequate cardiorespiratory fitness**—were voiced as important for preventing injury
 - To address this, **structured, triathlon-specific training plans** implemented over several months were recommended by participants

Non-modifiable Risk Factors Theme:

- **Environmental conditions** (weather, water conditions, motor vehicle traffic) and **participant crowding** were prominently perceived
 - Continue to **address through triathlon competition rules & policy updates** (drafting rules, rolling swim starts, course design)



Discussion: Key Takeaways Continued

- **Previous injury** is a well-established risk factor for future injury. To address this, participants recommended:
 - **Undergoing a pre-race medical exam** (also helpful for cardiovascular health screening)
 - **Seeking medical attention** (as needed) to mitigate the risk of future injury or reinjury
- **Having Prior Race Experience** was perceived as a preventive factor while **older age** was perceived as an injury risk factor
 - Previous literature has shown mixed results regarding these two findings
 - Triathlon experience may convey protective benefits while overall exposure to injury risk accumulates through successive years of chronic physical stress

Discussion: Key Takeaways Continued

- **Nutrition, The “Fourth” Discipline**

- **Optimizing nutrition/hydration strategies—before, during, and after competition—provides a significant opportunity to reduce triathlon-related health issues**
- Nutrition/hydration was perceived as critical for both injury prevention and injury risk
- During an Ironman, triathletes can consume ~3–4,000 kcal and expend ~10–11,000 kcal

- **Warning Signs (“Red flags”) Theme:**

- **“Red flag” symptoms are all symptoms associated with nutrition-related issues**
 - ✓ Exercise-induced hyponatremia, carbohydrate depletion, dehydration



Discussion: Death in Triathlon

Multiple study participants reported witnessing at least one triathlon-related death from drowning, cardiac emergency, or cycling collision.

- 135 race-related deaths in U.S. alone from 1985 to 2016
 - No clear underlying factors identified
 - Cardiovascular abnormalities were found in ~44%
- **Sudden cardiac death is the leading cause of death in athlete populations**
 - Pre-existing conditions (hypertrophic cardiomyopathy)
 - Athlete's Heart
 - Associated risk factors: older age, male, African American/black ethnicity
- Pre-race medical exams are recommended but not required by most race organizations
 - Cardiovascular screening (e.g., ECG) typically is not included
 - **Pre-race medical exams with cardiovascular screening could reduce risk to athletes**

Past & Future Research

Triathlon has evolved considerably over the years but there's still work to do!

- Changes to swim starts (rolling/staggered starts vs. mass start)
- Wetsuit guidelines
- Water safety personnel standards
- Drafting rules
- Standards for medical staff to athlete ratios

TriSafe triathlete recruitment/data collection is ongoing

- Cardiovascular health data
- Sleep and jetlag data (circadian disruption)
- Geomapping/weather data
- **More qualitative data, including on the mental aspect and female-specific topics!**
- Expanding study locations (829 US-sanctioned events, ~150 Ironman races annually)

TriSafe Team

Thank you to our triathlete participants!

Dr. Chris Connolly, PhD

Dr. Amanda Lamp, PhD

Dr. Erin Griffin, PhD

Dr. Thomas Gooding, PhD

Dr. Alex Montoye, PhD

Dr. Amy Bender, PhD

Dr. Sandhya Venugopal, MD

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Danny Smith

Questions?

Let's connect!

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